

PocketSmart – Validate Gemini API Connectivity

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1. Topic

Validating the Gemini API connectivity for PocketSmart

Epic: 1 – Gemini AI Initialization

2. Project Overview

PocketSmart is an AI-powered app that helps users track spending, set budgets, and get personalized money-saving suggestions. It uses Google Gemini AI to analyze spending data, so the connection to the Gemini API must be confirmed as working before the AI features are built.

3. Objective

To confirm that the backend can connect to the Gemini API using the secure API key, receive a valid response to a test prompt, and handle connection errors clearly.

4. Work Steps

- 1 Confirm that the Gemini API is enabled and the API key is stored in the .env file.
- 2 Install the Google Gemini Python library and the library that loads the .env file.
- 3 Load the API key from the environment and check that it is not empty.
- 4 Initialize the Gemini model in the backend.
- 5 Send a simple test prompt and check that a valid text response is returned.
- 6 Test an invalid or missing API key and confirm a clear error message is shown.
- 7 Test the request limit error and confirm the app handles it without crashing.
- 8 Record the test results and report the connectivity status to the team.

5. Main Parts of the Validation

- API key loading and security check
- Gemini model initialization
- Test prompt and response check
- Invalid key and missing key testing
- Request limit error handling
- Test result recording and reporting